1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 synthesis highlights a paradox: some tumors with apparent immune infiltration may nonetheless fail to benefit from immunotherapy. The ontologist points to a CCND1→GLS metabolic axis (with CXCR4 links) and EMT/Wnt–driven invasion as intersecting mechanisms that could render immune-rich tumors functionally immunosuppressed. 2) Tier 1 paths used: (a) CCND1 overexpression regulates GLS (direct metabolic reprogramming) and (b) GLS activity is linked to reduced immunotherapy benefit; these provide a tight mechanistic chain linking an oncogenic driver to a metabolic state that plausibly blunts anti-tumor immunity. I built on CCND1→GLS because it gives a manipulable metabolic node. 3) Tier 2 evidence: the CXCR4–GLS association supports how metabolism couples to trafficking/homing and microenvironment interactions that can protect tumor niches; EMT/Wnt support the aggressive, dissemination-prone phenotype that may coexist with metabolic immune suppression. These reinforce the mechanism that metabolism (GLS) conditions immune efficacy in CCND1-driven tumors. 4) Endpoint choice rationale: I chose therapy_response (immunotherapy response) because Core Path 3 directly implicates GLS in modulating benefit from immunotherapy, making this both mechanistically focused and clinically actionable. 5) Deliberate exclusions: I excluded transporter/docetaxel/nedaplatin and broader chemotherapy edges as they are tangential to the central immunometabolic puzzle; I also excluded listing multiple EMT/Wnt TFs or other metabolic enzymes in the core hypothesis to preserve maximal focus (they are placed in secondary_modifiers). I selected 'Tumor-Infiltrating Lymphocyte (TIL) density' as the imaging feature because it proxies immune presence on H&E and is the most relevant imaging correlate for immunotherapy response?

Working Hypothesis

Primary hypothesis: In TIL-rich mantle cell lymphoma (MCL), CCND1 amplification/overexpression drives upregulation of GLS, reprogramming tumor glutamine metabolism to create an immunosuppressive microenvironment that impairs effector T-cell function and antigen presentation, thereby reducing response to immunotherapy. A key testable prediction is that among TIL-rich tumors, those with high CCND1 and high GLS expression will have poorer immunotherapy responses than those with high CCND1 but low GLS expression (GLS-low), independent of TIL density. A parallel prediction is that pharmacological GLS inhibition or genetic disruption of the CCND1–GLS axis will restore or enhance immunotherapy sensitivity in CCND1/GLS-high, TIL-rich models. Secondary modifiers (CXCR4, CTNNB1, ZEB1) may modulate the magnitude of this effect by shaping the tumor microenvironment and invasion but are not essential for the core mechanism. This framework implies a testable immunometabolic axis: CCND1 → GLS → glutamine flux/redox state → TIL function → immunotherapy outcome.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CCND1, GLS
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CCND1: CCND1 overexpression regulates GLS (direct metabolic reprogramming) and (b) GLS activity is linked to reduced immunotherapy benefit; these provide a tight mechanistic chain linking an oncogenic driver to a metabolic state that plausibly blunts anti-tumor immunity.

GLS: GLS association supports how metabolism couples to trafficking/homing and microenvironment interactions that can protect tumor niches; EMT/Wnt support the aggressive, dissemination-prone phenotype that may coexist with metabolic immune suppression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CXCR4	Supporting modifier from knowledge graph
Secondary	CTNNB1	Supporting modifier from knowledge graph
Secondary	ZEB1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of MCL patient datasets to correlate CCND1/GLS expression and TIL metrics with immunotherapy outcomes;

Prediction 2: in vitro coculture of CCND1-high MCL cell lines with human T cells to assess T cell activation, proliferation, cytotoxicity, and antigen presentation with GLS manipulation (pharmacologic inhibition or gene knockdown);

Prediction 3: metabolic profiling (glutamine/glutamate flux, Seahorse assays, redox state) to link GLS activity to immunometabolic changes;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CXCR4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTNNB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.